

FRAMEWORK FOR TRUST

TRUST, ANCHORED TO EARTH

A protocol whitepaper

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Polygon Amoy Testnet

Table of Contents

1. Abstract	3
2. The Problem	3
3. Architecture: Four Primitives	3
4. System Design	4
5. Trust Scoring Model	5
6. Current Status	5
7. Proof of Existence	6
8. How FfT Differs	6
9. Roadmap	6
10. Open Research Questions	7
11. Licensing & Governance	7
12. Disclaimer	8
13. Conclusion	8

1. Abstract

Framework for Trust (FfT) is a decentralized protocol for verifying and scoring real-world events. It anchors physical-world occurrences — accidents, outages, floods, local disruptions — to fixed geographic coordinates on a public blockchain, and evaluates the trustworthiness of each report through a combination of on-chain reputation, semantic consistency analysis, and economic staking.

Unlike closed platforms that centralize location data and unilaterally decide what counts as “verified,” FfT is built as public infrastructure: the grid, the events, and the scoring logic are open, inspectable, and independent of any single operator. The system is currently live on the Polygon Amoy testnet, with a deployed contract suite, a working reputation and reward pipeline, and an indexed global grid of geographic quadrants.

This document describes the problem FfT addresses, its architecture, its trust-scoring model, its current development status, and the open research questions the project intends to pursue going forward. It intentionally avoids token-price speculation or investment framing: GCD, the protocol's utility token, is described here strictly in terms of its function within the system, not as a financial instrument.

2. The Problem

Every day, millions of real-world events happen around us: traffic incidents, power outages, floods, protests, local disruptions. This information exists, but it is fragmented — scattered across closed apps, social media threads, and municipal systems that don't talk to each other.

Three structural gaps make this data hard to trust and hard to use:

- No shared anchor. The same physical location is described differently by every platform that reports on it, with no common reference frame.
- No independent verification. Trust in a report typically depends on trusting the platform that hosts it, rather than on any property of the report itself.
- No persistent accountability. A false or misleading report costs its author nothing once removed — there is no lasting record tying reputation to accuracy.

FfT addresses all three by anchoring events to a public, permanent geographic grid; scoring reports using a transparent, on-chain-verifiable mechanism; and tying reputation to a token that can be earned through accurate contributions and lost through confirmed inaccuracy.

3. Architecture: Four Primitives

The protocol is built from four composable primitives. Each is intentionally minimal, so the system's behavior can be reasoned about from first principles rather than from implementation detail.

3.1 Geo-Quadrants

The Earth's land surface is divided into a fixed grid of $10^\circ \times 10^\circ$ cells, each represented as an ERC-721 NFT (GeoQuadrants contract). A quadrant is a permanent coordinate anchor: every trust event references the

quadrant it occurred in, giving the system a consistent spatial index independent of any mapping provider. The full land-only grid — approximately 325 quadrants, including both poles as special cases — is minted and live on testnet.

3.2 Trust Events

A trust event is a report of something that happened at a specific quadrant and time. Each event is signed by its submitter's wallet (EIP-191), giving it a cryptographic author and making it tamper-evident. Events are stored off-chain for scale, with their scoring inputs and outcomes anchored back to the chain.

3.3 GCD Token

GeoChainData (GCD) is the protocol's ERC-20 utility token. It is minted to a contributor's balance when a report is scored as trustworthy (reward), and burned when a report is confirmed false (slash). GCD balance functions as an on-chain reputation signal: it is read directly by the scoring engine as one input into how much a given wallet's future reports should be trusted.

GCD is currently a testnet utility token with no assigned monetary value and no public sale. Its role in the system is functional, not speculative — this document makes no claims about its future market value.

3.4 Semantic Search

Reports are embedded into a 384-dimensional multilingual vector space (paraphrase-multilingual-MiniLM-L12-v2) and stored in a Qdrant vector database. This lets the scoring engine check whether a new report is semantically consistent with prior reports about the same location — and lets anyone query the event graph by meaning rather than by exact keyword or coordinate match.

4. System Design

4.1 Layers

The system is organized into four layers, each with a narrow responsibility:

- Frontend — React with a Leaflet map, MetaMask wallet integration, and an event submission form.
- Backend — Python / FastAPI service that handles authentication, trust scoring, and acts as the on-chain oracle for reward and slash transactions.
- Data layer — MySQL for events, ledger, and quadrant indexing; Qdrant for semantic embeddings.
- Blockchain layer — four Solidity contracts deployed on Polygon Amoy, described in §4.2.

4.2 Smart Contracts

All contracts are written in Solidity ^0.8.26 using OpenZeppelin v5 access-control primitives, and are currently deployed on Polygon Amoy (chain ID 80002).

Contract	Address	Role
GeoChainData (GCD)	0x3131AcA746B7613390DED61613E5C0Ae9944B635	ERC-20 utility / reputation token

Contract	Address	Role
GeoQuadrants (NFT)	0x1421C0dd6D962fb5c5A29340C74bEE66AdA60BFb	ERC-721 geographic grid
FfTRegistry	0x58baE811a67E312BBB7B7fF1A438a2A0137155cd	On-chain contract address registry
FfTStaking	0x49568b041FD6F77dAD1611978043Ba2b18D84b92	Stake, reward, and slash logic

All four contracts share access-control roles (MINTER_ROLE, BURN_ROLE, ORACLE_ROLE) that gate who may mint, burn, or trigger reward/slash actions — currently held by the protocol's backend oracle wallet during the testnet phase.

5. Trust Scoring Model

Every submitted event moves through the same four-stage pipeline:

STAGE 1 – SIGN

The contributor signs the event payload with their wallet (EIP-191). This is cryptographic proof that a specific address made a specific, unaltered claim.

STAGE 2 – SCORE

The backend computes a trust score from four inputs: the contributor's on-chain GCD reputation, semantic similarity to prior events at the same quadrant, anti-Sybil cluster detection across the embedding space, and the contributor's staked amount, if any.

STAGE 3 – REWARD

If the event clears the scoring threshold, GCD.reward() mints tokens to the contributor's on-chain balance — a real, verifiable transaction, not just a database entry.

STAGE 4 – SLASH

If an event is later confirmed false, GCD.burn() destroys tokens from the offending balance, creating a permanent, public record of the infraction.

This pipeline is deliberately symmetric: the same mechanism that rewards accurate contribution also penalizes inaccurate contribution, so reputation is earned continuously rather than granted once.

6. Current Status

FfT is live on Polygon Amoy testnet, not a specification without an implementation. As of this draft:

- All four contracts are deployed, role-configured, and verified end-to-end with a real signed transaction (reward flow confirmed on-chain).
- Approximately 325 land quadrants — the full 10°×10° land grid, including both poles — are minted and indexed.
- The semantic search layer is running on real multilingual embeddings, not a placeholder.
- The frontend supports wallet connection, signed event submission, and live map visualization of the quadrant grid.

Testnet core: ~95% complete Public beta readiness: ~85% complete

The project is developed and maintained by a small team of independent developers. This is stated plainly rather than obscured: it explains both the coherence of the current architecture and the pace at which new capabilities are added.

7. Proof of Existence

To establish an independently verifiable record of authorship and timing, the project's core documents are anchored to the Bitcoin blockchain via OpenTimestamps — a proof that does not rely on trusting FfT's own claims.

Bitcoin block: 958276

Attested as of: 2026-07-16 (CET)

Documents anchored: README.md, this whitepaper draft

Anyone can verify this claim independently at opentimestamps.org, without needing to trust the project's authors — the proof is validated against the Bitcoin blockchain itself.

8. How FfT Differs

FfT is not the first attempt to bring verifiability to physical-world data. It is worth being precise about how it differs from adjacent approaches, without framing this as a competitive attack:

- Versus centralized reporting apps: FfT does not require trusting a single company's moderation decisions — reputation and scoring logic are on-chain and inspectable.
- Versus generic DePIN sensor networks: FfT's unit of record is a human- or device-submitted event tied to a geographic quadrant, not a continuous sensor stream — it is designed for discrete, describable occurrences, not telemetry.
- Versus reputation systems built on off-chain databases: FfT's reward and slash actions are real, signed blockchain transactions, not entries in a private ledger that could be altered without a trace.

9. Roadmap

PHASE A – Soft Public Release (in progress)

Public GitHub repository, documentation, and this whitepaper made available to early collaborators and the wider DePIN community.

PHASE B – Community Feedback

Open review of the architecture and trust model, incorporating outside critique before any mainnet commitment.

PHASE C – Academic & Research Collaboration

Formal engagement with researchers on the open questions outlined in §10, particularly around provable model execution and dispute resolution.

PHASE D – Mainnet Path

A mainnet deployment is a future goal, not a current commitment. It will follow, not precede, external review of the trust model and contract security.

10. Open Research Questions

FfT is transparent about what remains unresolved. These are not marketing gaps to be waved away — they are the project's actual research frontier, and an open invitation for collaboration:

10.1 Atomic Settlement

Reward and slash actions currently rely on a single backend oracle wallet to broadcast on-chain transactions. A more decentralized design would settle scoring outcomes atomically, without a single point of trust or failure in the oracle layer.

10.2 Model Execution Proof

The semantic scoring model runs off-chain. There is currently no cryptographic proof that a given trust score was actually produced by the stated model on the stated input — an open problem shared with most AI-plus-blockchain systems, and one FfT intends to address rather than ignore.

10.3 Post-Quantum Cryptography

EIP-191 signatures, like most current blockchain cryptography, are not post-quantum secure. FfT's event-signing layer will need a migration path as post-quantum standards mature.

10.4 Dispute Resolution

The current slash mechanism assumes a moderator can confidently mark an event as false. A more robust system needs a formal, adversarial dispute process — closer to on-chain arbitration than unilateral moderation.

Researchers or collaborators interested in any of these questions are welcome to engage through the project's public repository.

11. Licensing & Governance

FfT's source code is released under the license specified in the project's public repository (see the LICENSE file). The project is currently maintained by a small team of independent developers; governance is expected to evolve as the contributor base grows, and any changes will be documented in the repository rather than asserted here in advance.

12. Disclaimer

This document is a technical and conceptual description of the Framework for Trust protocol. It is not a prospectus, an offer of securities, or investment advice. GCD is described solely in terms of its function within the protocol's reputation and reward mechanism. The system described here is deployed on a public testnet and remains under active development; nothing in this document should be read as a guarantee of future functionality, timelines, or value of any kind.

13. Conclusion

Framework for Trust is an attempt to give physical-world events the same properties that public blockchains gave financial transactions: a shared reference frame, independent verifiability, and persistent accountability. The architecture is live, not hypothetical — a deployed contract suite, a working scoring pipeline, and a fully indexed global grid back every claim made in this document.

What remains is not a secret roadmap but an open one: the questions in §10 are stated plainly because solving them in the open, with outside scrutiny, is more likely to produce something durable than solving them alone.